Dynamic Planet C - 2023 HDI Dynamic Planet C - High Desert Invitational 2023 - 02-04-2023
Answer Keys for Grading (/hdso/SM/AnswerKeys?tid=000D0R)

Read questions carefully. You will not be penalized by typos.

Be sure this test is the ONLY BROWSER open NO other Browsers can be open for ANY REASON

Notes above MUST be printed as a HARD PAPER COPY NOT DIGITAL

1. (10.00 pts)
"The Science Olympiad Academic Honesty Satellite Policy":

1. Entering browser(s) outside this "scilympiad" Test website for more than 60 seconds WILL earn penalty deductions to test score and/or be disqualified from this event.

2. Only resources indicated in the rules for this event may be used during the test.

3. Inappropriate behavior in the chat box WILL NOT be tolerated and will earn penalty deductions to test score and/or be disqualified from this event.

- (●) A) I will abide by Science Olympiad Academic Honesty Policy

2. (4.00 pts) Which of the following is the sinuosity of a river?

(Mark **ALL** correct answers)
- [x] A) actual path length/shortest path length
- [ ] B) sin(length of channel/width of channel)
- [ ] C) the number of meanders in a given length of the river
- [x] D) channel length/valley length

3. (4.00 pts) Which of the following is true of porosity?

(Mark **ALL** correct answers)
- [x] A) It depends on grain size of sediments
- [x] B) Poorly sorted sediments tend to have a higher porosity
- [x] C) It determines how much water a rock or sediment can contain
- [x] D) The measurement of porosity ranges from 0-1

4. (4.00 pts) What is true of suspended load?

(Mark **ALL** correct answers)
- [ ] A) This is used to describe the dissolved ions that are carried in solution in a stream or river.
- [x] B) This is used to describe the sediment that is carried by the flow of a river and almost never touches the river bed.
- [x] C) It is generally comprised of particles of fine sand, silt, or clay
- [ ] D) It is generally comprised of Na^+, K^+, and PO_4^{3-}

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5. (4.00 pts) What is the force between groundwater and the surface of rocks/sediment (due to ions on the surface of mineral grains)?

(Mark **ALL** correct answers)

- [ ] A) capillarity
- [ ] B) London dispersion forces
- [ ] C) capillary tension
- [x] D) force of molecular attraction

6. (4.00 pts) The following are often found in rocks that form karst formations:

(Mark **ALL** correct answers)

- [ ] A) NaCl
- [ ] B) $CaSO_4$
- [x] C) $CaCO_3$
- [ ] D) NH_4

7. (4.00 pts) 7. Where is the water fastest in a meandering stream?

(Mark **ALL** correct answers)

- [ ] A) At the surface of the water
- [x] B) Right below the surface of the water
- [ ] C) Along the sides of the channel
- [ ] D) At the very bottom of the channel

8. (4.00 pts) Which of the following occur when a lake undergoes turnover?

(Mark **ALL** correct answers)

- [x] A) The water throughout the lake mixes
- [ ] B) The water at the very bottom of the lake remains dense and cold
- [x] C) The water becomes a uniform temperature
- [x] D) Thermocline disappears

9. (4.00 pts) Stream capacity is dependent on:

(Mark **ALL** correct answers)

- [ ] A) Type of rock that composes the river channel
- [ ] B) Volume of the river
- [ ] C) Wetted perimeter of the river channel
- [x] D) Velocity of the river

10. (4.00 pts) The following are all examples of nick points except:

(Mark **ALL** correct answers)

- [ ] A) lake

- [ ] B) cascade waterfall
- [x] C) fault
- [ ] D) fan waterfall

11. (1.00 pts) As the velocity of a stream decreases, which transported particle size would settle to the stream's bottom first?

- ○ A) clay
- ○ B) silt
- ◉ C) pebble
- ○ D) sand

12. (1.00 pts) Which or the following is not an environmental function of wetlands?

- ○ A) increasing runoff
- ○ B) absorbing and removing pollutants from water
- ◉ C) trapping carbon that would otherwise be released into the air
- ○ D) controlling floods

13. (1.00 pts) The terms subsidence refers to

- ○ A) failure of the groundwater supply
- ○ B) accumulation of sill behind a dam
- ◉ C) sinking of ground when water has been withdrawn
- ○ D) intrusion of salt water into a freshwater aquifer

14. (1.00 pts) A common feature of thermal pollution and artificial eutrophication is that they both

- ○ A) have sources that are difficult to identify and control.
- ○ B) cause large mats or algae to bloom in fresh water.
- ○ C) are a result or power plants and other industrial activity.
- ◉ D) decrease the amount of oxygen dissolved in water

15. (1.00 pts) Karst landscapes occur in areas high in

- ○ A) calcium
- ◉ B) limestone
- ○ C) shale
- ○ D) sandstone

16. (1.00 pts) Which of these landforms is characteristic of an area with karst topography?

- ○ A) mountains

B) canyons

C) sinkholes

D) drumlins

17. (1.00 pts) Four of the following are harmful environmental effects of a severe drought, one is not. Choose the one that is not.

A) dries out soil

B) reduces moisture in the air from evaporation

C) reduces stream flows

D) decreases tree growth and biomass

E) reduces crop yields

18. (1.00 pts) Ocean water _____ than fresh water.

A) freezes at a higher temperature

B) contains more salts

C) supports fewer fish

D) is less abundant

19. (1.00 pts) Which of the following statements is not correct?

A) Lakes may contain a region that receives little sunlight.

B) Wetlands help remove Pollutants from water.

C) Swamps are dominated by woody shrubs and water-loving trees.

D) Rivers generally move faster, and their oxygen levels decrease, as they near the ocean

20. (1.00 pts) Which factor least affects the rate of runoff?

A) slope

B) vegetation

C) volume of runoff

D) nearness to water

21. (4.00 pts) Which of the following terms are associated with transportation?

(Mark **ALL** correct answers)

A) Saltation

B) Solution

C) Corrosion

D) Attrition

E) Traction

F) Suspension

22. (1.00 pts) What will occur if the precipitation rate is greater than the initial infiltration capacity?

- (●) A) The precipitation will immediately pool at the surface.
- ( ) B) The precipitation will immediately infiltrate into the ground
- ( ) C) The precipitation will infiltrate, but some precipitation will remain on the land surface.
- ( ) D) The precipitation will never infiltrate into the ground.

23. (1.00 pts) The respective depth and width of a rectangular channel are 4 meters and 10 meters.

Calculate the discharge of water through the channel if the bed slope is 1/600 and Chezy's constant is 50.

_____ m^3

- ( ) A) 3.04
- ( ) B) 3.33
- (●) C) 121.66
- ( ) D) 145.99

24. (1.00 pts) Streams with which characteristic are likely to have the most biodiversity?

- ( ) A) Strahler order of 0
- ( ) B) Lower Strahler order
- (●) C) Higher Strahler order
- ( ) D) Strahler orders have no impact on biodiversity

25. (1.00 pts) Which of the following is not a common pollutant associated with point source pollution?

- ( ) A) Carbon dioxide
- ( ) B) Heavy metals
- ( ) C) Nitrogen oxides
- (●) D) Ammonium nitrate

26. (1.00 pts) Between points A and B on a river, there is a river-parallel distance of 1.08 km and a straight-line distance of 0.60 km, what is the sinuosity of this river?

- ( ) A) .48
- ( ) B) .56
- ( ) C) 1.68
- (●) D) 1.8

27. (1.00 pts) Which type of rock would you expect to find a cavern in?

- (●) A) Granite
- ( ) B) Shale
- ( ) C) Both

D) Neither

28. (1.00 pts) On the outside of a meander curve, when compared to the inside, the river's velocity is _____.

A) Equal
B) Lower
C) higher
D) variable

29. (1.00 pts) What identifies the "competence" of a stream?

A) Stream velocity
B) The maximum particle size the stream carries
C) The volume of water in a particular section of a stream
D) The quantity of sediment carried by a stream

30. (1.00 pts) What is the name of the device used to measure evaporation?

A) Turbidimeter
B) Hygrometer
C) Atmometer
D) Lysimeter

31. (1.00 pts) How are oxbow lakes formed?

A) Tectonic activity
B) Erosion and deposition
C) High stream velocity
D) Flooding
E) Glaciers

32. (1.00 pts) Where could one find a knickpoint?

A) Niagara Falls on the Canada-United States border
B) Bridalveil Fall in Yosemite
C) Dry Falls in Washington
D) All of the above
E) None of the above

33. (1.00 pts) What direction do knickpoints migrate, why?

A) Upstream due to erosion

B) Upstream due to sediment deposition

C) Downstream due to erosion

D) Downstream due to sediment deposition

34. (1.00 pts) Which of the following sediments forms the most effective aquiclude?

A) Sand

B) Clay

C) Gravel

D) Sandstone

35. (1.00 pts) Which of the following causes river rejuvenation?

A) Rise in sea level

B) Uplift

C) Heavy Rain

D) Bifurcation

36. (1.00 pts) Which of the following is not a form of accordant drainage?

A) Consequent

B) Antecedent

C) Resequent

D) Subsequent

E) Insequent

37. (1.00 pts) Which of the following is not a zone of a stream?

A) Fluvial zone

B) Deposition zone

C) Headwater zone

D) Transfer zone

38. (2.00 pts) Which of the following contribute to the formation of sinkholes? Select all that apply:

(Mark **ALL** correct answers)

A) Excess withdrawal of groundwater

B) Cave roof collapses

C) Limestone bedrock erosion

D) Water table lowers

39. (1.00 pts) What is the maximum salinity in freshwater?

A) 10 ppm
B) 100 ppm
C) 500 ppm
D) 1000 ppm

40. (1.00 pts) What causes the formation of knickpoints?

A) Isostatic rebound
B) Tectonic uplifting
C) Decreased ocean water influx
D) All of the above

41. (1.00 pts) What type of sediment load is deposited first when a river enters the ocean?

A) Dissolved load
B) Bed load
C) Suspended load
D) None of the above

42. (1.00 pts) What desalination method uses high pressure to force saltwater through a membrane filter?

A) Diffusion
B) Distillation
C) Active Transport
D) Reverse osmosis

43. (1.00 pts) Which of these organisms do not live in a freshwater environment?

A) Mayfly
B) Asian carp
C) Zooxanthellae
D) Water lettuce

44. (1.00 pts) Which of the following wetlands is known for its lack of nutrients and inability to support large plantlife?

A) Fens
B) Swamps
C) Marshes
D) Bogs

45. (1.00 pts) Which of the following effects does sediment have on aquatic organisms?

A) Increasing small-order stream productivity

B) Habitat destruction

C) Increasing biodiversity

D) Reducing competition among aggressive species

46. (2.00 pts) The emergence of groundwater at the land surface, usually at a clearly defined point is known as _____

spring

47. (2.00 pts) The equation describing the flow of a fluid through a porous medium is known as _____ 's Law .

Darcy

48. (2.00 pts) A stream that receives groundwater discharge is known as a(n) _____ stream.

graining or effluent

49. (2.00 pts) Pure water reaches its maximum density at _____ º centigrade.

4

50. (2.00 pts) Stream flow in which all water molecules travel in roughly parallel paths is known as _____ flow.

laminar

51. (4.00 pts) A valley has a lot in common with a basin; the difference is a _____ has an outlet to drain the water, while a _____ doesn't.

valley

basin

52. (2.00 pts) _____ valleys are usually characterized by V shape cross-sections and steep slopes.

River

53. (2.00 pts) The Nile River Valley is a classic example of a _____ valley.

flat-floored

54. (2.00 pts) _____ valleys are formed when tectonic plates underneath the earth's surface expand the crust.

Rift

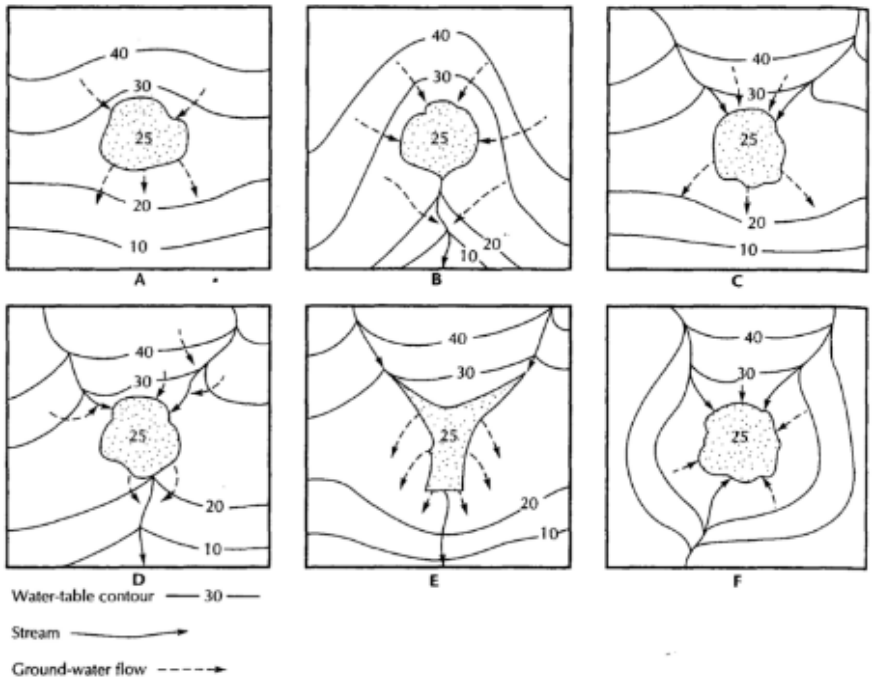
55. (6.00 pts)
Assuming Lincoln Creek runs 10,000 ft in distance (not displacement), is the approximate gradient of Lincoln Creek shown is ____%?

Assume Lincoln Creek runs 9,000 ft in displacement only (straight-line distance) and averages 15 ft of width and 3 ft of depth. Calculate its sinuosity. _____(100th place)

What is the Strahler order number of Lincoln Creek? _____ (digit)

4-6% accepted	1.11	3

56. (6.00 pts)



A. Which of the letters represents a lake which is being fed only by groundwater and drained by stream flow? _____

B. Which of the letters represents a reservoir created behind a dam? _____

The water level created by this reservoir is _____ (less or greater) than the adjacent ground-water level.

B	E	greater

57. (2.00 pts) The volcanoes originally start out with completely flat slopes of lava but over time, valleys are carved into the sides by streams. What is the name of this process?

downcutting

58. (2.00 pts) The Manoa stream has a sinuosity of 1.67. What type of stream is the Manoa stream? _____ stream

Meandering

59. (2.00 pts) An _____ lake is generally warm, stratified, and mixes sporadically.

oligomictic

60. (2.00 pts) _____is the name of year round unfrozen ground, usually found underneath permafrost?

Talik or active layer

61. (2.00 pts) On the drive to Horseshoe bend you drive past many dry stream beds.

These beds have what type of stream in them?

Ephemeral

62. (6.00 pts) What are the 3 common/not so common names that a dry stream bed is called? (ENTER ALPHABETICALLY)

arroyo | Dry wash | wadi

63. (2.00 pts) _____ zone is another name for the aerated zone?

Vadose

64. (6.00 pts) Order the epilimnion, hypolimnion, and metalimnion from the top of the lake to the bottom.

epilimnion | metalimnion | hypolimnion

65. (6.00 pts) Determine whether it is a point source or nonpoint source of pollution. (TYPE ONLY APPROPRIATE UNDERLINED WORD)

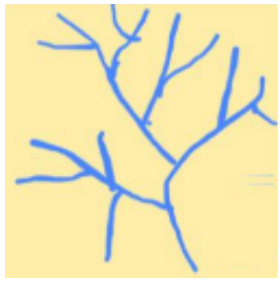
A. Agricultural runoff

B. Wastewater treatment pipes

C. Rain with chemicals

nonpoint | point | nonpoint

66. (2.00 pts)



This diagram is an example of a _____ drainage system.

Dendritic

67. (2.00 pts) What rock makes up Karst landscapes?

limestone

68. (6.00 pts) Identify the following waterfalls with their appropriate names:



A B C

A _____ B _____ C_____

multistep | plunge | Cascade

69. (6.00 pts) Identify the following waterfalls with their appropriate names:



D E F

D _____ E _____ F _____

punchbowl | cataract | fan

70. (6.00 pts) How does a lake form from a glacier?

Expected Answer: Near the end of the last glacial period, glaciers began to retreat. A retreating glacier often left behind large deposits of ice (4 point) in hollows between drumlins or hills. (4 point) As the ice age ended, these melted to create lakes. (4 point) Add another point if no points missed up to this point.

71. (6.00 pts) What is the role of the stagnation zone beneath a lake?

Expected Answer: In seepage lakes that are surrounded by the water table on all sides, the stagnation zone leads to upward seepage through the lake bottom. As long as the stagnation zone is present, the lake will never lose water through the bottom. (Answer just needs to include: that the lake will never lose water through the bottom OR leads to upward seepage through the lake bottom

72. (6.00 pts) Define “critical zone”.

Expected Answer: The thin layer on top of the Earth's crust where air, water, soil, and rock interact to make life possible.

73. (6.00 pts) What type of lake will Horseshoe Bend become over time? How are these lakes formed?

Expected Answer: Oxbow lake (5 point). A meander in a stream is cut off by the stream seeking the easiest and fastest way downhill forming that oxbow lake (5 points).

74. (6.00 pts) Why is it difficult to farm in an area of Karst topography, even if soil quality is good and rainfall is adequate?

Expected Answer: The large voids and fractures in the rock allows water to drain very quickly away from the surface so it is difficult to access for farming. (5 points. Kinda all or nothing)

75. (6.00 pts) The walls above the river prevent water from exiting, what type of floodplain feature could this be classified as?

Expected Answer: levee

76. (6.00 pts) What differentiates aquifers, aquitards, aquicludes, and aquifuges?

Describe the sediment forms that characterize each of these geological formations

Expected Answer: 3 points each: Aquifers are permeable formations that can both store water and yield water, made up of sand, gravel, etc. Aquitards are partially permeable formations that can store water and slowly yield it, made up of materials like sandy clay. Aquicludes are impermeable formations that can store water but not yield it, made up of hard clays. Aquifuges are impermeable formations that can neither store water nor yield it, made up of compact rocks such as basalt or granite.

CONGRATULATIONS you have reached the END of the TEST

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